

Algorithm Analysis And Data Structures

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Chapter 1: INTRODUCTION

1.1. What's the Book About?

1.2. Mathematics Review

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1.2.2. Logarithms

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1.2.4. Modular Arithmetic

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1.3. A Brief Introduction to Recursion

Recursion and Induction

1.1. What's the Book About?

Algorithm Analysis
and Data Structures

What is an Algorithm?

An algorithm is

- A set of logical steps to accomplish a task.
- A “recipe of action”.
- A way of describing *behavior*.

Problems:

■ Selection Problem:

- Find the k^{th} -largest number from a group of N numbers

■ Puzzle

	1	2	3	4
1	t	h	i	s
2	w	a	t	s
3	o	a	h	g
4	f	g	d	t

Features of the used-Hardware & Software have to be the same.

1- Size: $n*m$

2- Dictionary (Number of Words)

3- Problem Description

4- Problem Solving

(Algorithms, Methods, ...)

Figure 1.1 Sample Word Puzzle

1.2. Mathematics Review

Reminding of Basic Formulas in
Mathematics to be able to derive or
reviews basic **PROOF** techniques

1.2.1. Exponents

- $X^A * X^B = X^{A+B}$
- $X^A / X^B = X^{A-B}$
- $(X^A)^B = X^{A*B}$
- $X^A + X^A = 2 * X^A \neq X^{2*A}$
- $2^N + 2^N = 2 * 2^N = 2^{N+1}$
- $1 + 2 + 4 + \dots + 2^N = 2^{N+1} - 1$
- ...

1.2.2. Logarithms

All logarithms are to the base 2, unless otherwise specified

- **Definition:** $\log_x A = B$ is equivalent to $X^B = A$

Theorem 1.1 :

$$\log_A B = \log_C B / \log_C A \quad ; \quad A, B, C > 0, A \neq 1$$

1.2.2. Logarithms

Proof:

Let $X = \log_C B$, $Y = \log_C A$ and $Z = \log_A B$

Then $C^X = B$, $C^Y = A$, $A^Z = B$

combining these three equalities yields

$$(C^Y)^Z = C^X = B$$

Therefore $X = Y \cdot Z$ which implies

$$Z = X/Y \rightarrow \log_A B = \log_C B / \log_C A$$

1.2.2. Logarithms

Theorem 1.2: $\log(A*B) = \log(A) + \log(B)$; $A, B > 0$

■ **Proof:**

- Let $X = \log(A)$ and $Y = \log(B)$
- Hence $2^X = A$ and $2^Y = B$
- Combining the above, $A*B = 2^X * 2^Y = 2^{X+Y}$
- Implies, $\log(A*B) = X + Y = \log(A) + \log(B)$
- Hence proved!

1.2.2. Logarithms

Other useful formulas:

- $\log(A/B) = \log(A) - \log(B)$
- $\log(AB) = \log(A) + \log(B)$
- $\log(X) < X$ for all $X > 0$
- $\log(1)=0, \log(2)=1, \log(1,024)=10,$
- $\log(1,048,576)=20$

1.2.3. Series

Geometric Series

- $\sum_{i=0..N} 2^i = 2^{N+1} - 1$

- $\sum_{i=0..N} A^i = (A^{N+1} - 1) / A - 1$

- If $0 < A < 1$, $\sum_{i=0..N} A^i \leq 1 / (1-A)$

- If $N \rightarrow \infty$, $\sum_{i=0..N} A^i$ approaches $1 / (1-A)$

1.2.3. Series

Arithmetic Series

$$\blacksquare \sum_{i=1..N} i = \frac{N (N+1)}{2}$$

$$\blacksquare \sum_{i=1..N} i^2 = \frac{N (N+1)(2N+1)}{6}$$

1.2.3. Series

Arithmetic Series

- $\sum_{i=1..N} i = \frac{N(N+1)}{2}$
- $\sum_{i=1..N} i^2 = \frac{N(N+1)(2N+1)}{6}$
- $\sum_{i=1..N} i^k \cong \frac{N^{k+1}}{|k+1|} \quad (k \neq -1)$

1.2.3. Series

Harmonic Numbers:

$$HN = \sum_{i=1..N} \frac{1}{i} \approx \log_e N \quad \text{Error} = 0.57721566$$

Two General Formulas:

- $\sum_{i=1..N} F(i) = N * F(N)$
- $\sum_{i=k..N} F(i) = \sum_{i=1..N} F(i) - \sum_{i=1..k-1} F(i)$

1.2.4. Modular Arithmetic

We say that *A is congruent to B modulo N*,

Written as $A \equiv B \pmod{N}$, if N divides $(A-B)$

In other words, the remainder is the same if either A or B are divided by N

- $81 \equiv 61 \equiv 1 \pmod{10}$

Similarly,

- $A+C \equiv B+C \pmod{N}$ and $AD \equiv BD \pmod{N}$

1.2.5. The P Word

- Proof by Induction
- Proof by Counterexample
- Proof by Contradiction

Proof by Induction

- Given a theorem
- First prove a **base case**
 - Show the theorem is true for some small degenerate values
- Next assume an **inductive hypothesis**
 - Assume the theorem is true for all cases up to some limit **k**
- Then prove that the theorem holds for the next value (**$k+1$**)

Proof by Induction

Example : Fibonacci Series

$$F_0 = 1, F_1 = 1, F_2 = 2, F_3 = 3 \dots F_i = F_{i-1} + F_{i-2}, \quad i > 1$$

- Show that
 - $F_i < (5/3)^i$, for $i \geq 1$
- Base case: $F_0 = 1 = F_1 < 5/3$
- Inductive Hypothesis
 - $F_i < (5/3)^i$, for $i = 1, 2, \dots, k$

Proof by Induction

Example 1: Fibonacci Series

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- Show that
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- Inductive Hypothesis
 - $F_i < (5/3)^i$, for $i = 1, 2, \dots, k$

Proof by Induction

- Now prove that $F_{k+1} < (5/3)^{k+1}$
- From definition of Fibonacci Sequence

$$F_{k+1} < F_k + F_{k-1}$$

- Using inductive Hypothesis

- $$\begin{aligned} F_{k+1} &< (5/3)^k + (5/3)^{k-1} \\ &< (5/3)^k + (5/3)^{k-1} \\ &< (5/3)^{k+1} [3/5 + (3/5)^2] \\ &< (5/3)^{k+1} [24/25] \\ &< (5/3)^{k+1} \end{aligned}$$

Proof by Induction

Example 2

PROOF:

$$\sum_{i=1}^{n+1} i^2 = \sum_{i=1}^n i^2 + (n+1)^2$$

Applying the inductive hypothesis, we obtain

$$\sum_{i=1}^{n+1} i^2 = \frac{n(n+1)(2n+1)}{6} + (n+1)^2$$

Proof by Induction

$$\begin{aligned}\sum_{i=1}^{n+1} i^2 &= \frac{n(n+1)(2n+1)}{6} + (n+1)^2 \\&= (n+1) \left[\frac{n(2n+1)}{6} + (n+1) \right] \\&= (n+1) \frac{2n^2 + 7n + 6}{6} \\&= \frac{(n+1)(n+2)(2n+3)}{6} \\&= \frac{(n+1)[(n+1)+1][2(n+1)+1]}{6}\end{aligned}$$

Proof by Counterexample

Theorem: The statement $F_k < k^2$ in the Fibonacci series is false

Proof: $F_{11} = 144 > 11^2$

Proof by Contradiction

Initially assume that the theorem is false, then show that some known property would be false as well

Example: “There are an infinite number of prime numbers”

Proof: Assume the theorem is false

- Let $P_1, P_2, \dots, P_k$ be all the primes in increasing order
- $N = P_1 P_2 \cdots P_k + 1$ is $> P_k$ and so it is not a prime.
But it is also not divisible by any of the listed primes, contradicting the factorization of integers into primes.

1.3. A Brief Introduction to Recursion

*A function that is defined in terms of itself is called **recursive**.*

Four fundamental rules of recursion:

- 1) Base Cases*
- 2) Making Progress*
- 3) Design Rule*
- 4) Compound Interest Rule (No duplicate work)*

Example : $F(0)= 0, F(x) = 2F(x -1)+x^2$

$F(0)= 0, F(1)= 1, F(2)= 6, F(3)= 21, \dots$

```
#include <stdio.h>

/* START: fig1_2.txt */
    int
    F( int X )
    {
/* 1*/        if( X == 0 )
/* 2*/            return 0;
        else
/* 3*/            return 2 * F( X - 1 ) + X * X;
    }
/* END */

main( )
{
    printf( "F(5) = %d\n", F( 5 ) );
    return 0;
}
```

Figure 1.2 A Recursive Function

```

#include <stdio.h>

/* START: fig1_3.txt */
    int
    Bad( unsigned int N )
    {
/* 1*/        if( N == 0 )
/* 2*/            return 0;
        else
/* 3*/            return Bad( N / 3 + 1 ) + N - 1;
    }
/* END */

main( )
{
    printf( "Bad is infinite recursion\n" );
    return 0;
}

```

Figure 1.3 Nonterminating Recursive Function

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Questions and Answers

Thank you ...

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